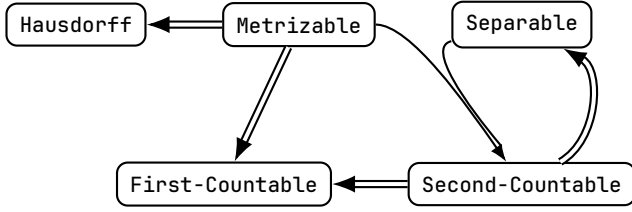


### 0.0.1 Metrizable Theorem Diagram



| Topology for $\mathbb{R}$ : $\mathcal{T}_{id} \subset \mathcal{T}_f \subset \mathcal{T}_u \subset \mathcal{T}_l \subset \mathcal{T}_d, \mathcal{T}_f \subset \mathcal{T}_c$ |                    |                 |                 |                 |                 |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Topology                                                                                                                                                                    | $\mathcal{T}_{id}$ | $\mathcal{T}_f$ | $\mathcal{T}_u$ | $\mathcal{T}_l$ | $\mathcal{T}_d$ | $\mathcal{T}_c$ |
| First-Countable                                                                                                                                                             | Yes                | No              | Yes             | Yes             | Yes             | No              |
| Second-Countable                                                                                                                                                            | Yes                | No              | Yes             | No              | No              | No              |
| Hausdorff                                                                                                                                                                   | No                 | No              | Yes             | Yes             | Yes             | No              |
| Metrizable                                                                                                                                                                  | No                 | No              | Yes             | No              | Yes             | No              |
| Separable                                                                                                                                                                   | Yes                | Yes             | Yes             | Yes             | No              | No              |
| Compactness                                                                                                                                                                 | Yes                | Yes             | No              | No              | No              | No              |

Let a Topological Space  $(X, \mathcal{T})$  be given. For  $X$ , the following theorems hold:

**Theorem 0.0.1.** If Metrizable, then Hausdorff.

*Proof.* Let  $(X, \mathcal{T})$  be a metrizable topological space. Then, there is a metric  $d: X \times X \rightarrow \mathbb{R}$ .

Let distinct two point  $x, y \in X$  be given. And, put  $\varepsilon = \frac{d(x, y)}{2} > 0$ .

Now,  $x \in B_d(x; \varepsilon)$  and  $y \in B_d(y; \varepsilon)$  but there intersection is empty set because:

Choose  $p \in B_d(x; \varepsilon) \cap B_d(y; \varepsilon)$ . Then,  $d(x, p), d(y, p) < \varepsilon$ .

But, this implies that

$$d(x, y) \leq d(x, p) + d(p, y) < 2\varepsilon = d(x, y)$$

Thus, contradiction. Consequently, given topological space satisfies Hausdorff property. □

**Lemma 0.0.2.** Let  $(X, d)$  be a Metric Space.

For any  $U \in \mathcal{T}_d$ ,  $x \in U$ , there is  $\delta > 0$  such that  $x \in B_d(x, \delta) \subset U$ .

*Proof.* Let  $U \in \mathcal{T}_d$  be given.

Then, there exist  $y \in X$  and  $\varepsilon > 0$  such that  $x \in B_d(y, \varepsilon) \subset U$ . For  $\delta = \varepsilon - d(x, y) > 0$ ,  $B_d(x, \delta) \subset B_d(y, \varepsilon)$  because:

$$\begin{aligned}
 p \in B_d(x, \delta) &\iff d(p, x) < \delta \\
 &\implies d(p, y) \leq d(p, x) + d(x, y) < \delta + d(x, y) = \varepsilon \\
 &\iff p \in B_d(y, \varepsilon)
 \end{aligned}$$

□

**Theorem 0.0.3.** If Metrizable, then First-Countable.

*Proof.* Let  $(X, \mathcal{T})$  be a metrizable topological space, and  $d$  be a metric that induces  $\mathcal{T}$ .

For any  $x \in X$ , define

$$\beta_x \stackrel{\text{def}}{=} \left\{ B_{\frac{1}{n}}(x) \mid n \in \mathbb{N} \right\}$$

Trivially,  $\beta_x$  is countable set. Want To Show:  $\beta_x$  be a local base of  $x$ .

**LB1)** is clear.

**LB2)** Let  $U \in \mathcal{T}$  with  $x \in U$  be given. Since above lemma, there is  $\delta > 0$  such that  $x \in B_d(x, \delta) \subset U$ .

Let  $n \in \mathbb{N}$  be a number such that  $1 < \delta n$ , then  $B_d\left(x, \frac{1}{n}\right) \in \beta_x$  satisfies

$$x \in B_d\left(x, \frac{1}{n}\right) \subset B_d(x, \delta) \subset U$$

□

**Theorem 0.0.4.** If Metrizable and Separable, then Second-Countable.

*Proof.* Since  $X$  is Separable, let  $A = \{x_i \mid i \in \mathbb{N}\}$  be a countable dense subset of  $X$ .  
Let  $d: X \times X \rightarrow \mathbb{R}$  be a metric on  $X$  that induces  $\mathcal{T}$ . Define

$$\beta \stackrel{\text{def}}{=} \left\{ B_d \left( x_i, \frac{1}{n} \right) \mid i, n \in \mathbb{N} \right\}$$

Clearly,  $\beta$  is Countable and  $\beta \subset \mathcal{T}$ .

Enough to Show that: For any  $U \in \mathcal{T}$ , there exists  $\mathcal{C} \subset \beta$  such that

$$U = \bigcup_{u \in \mathcal{C}} u$$

Let  $U \in \mathcal{T}$  be given. Since  $\mathcal{T}$  is consisted by open balls induced a metric  $d$ ,  
for any  $x \in U$ , there exists  $r > 0$  such that  $B_d(x, r) \subset U$  by the lemma.  
Set  $\delta_x = \frac{1}{2n}$  such that  $1 < nr$ . Since  $A$  is dense in  $X$ ,

$$A \cap B_d(x, \delta_x) \neq \emptyset$$

That is, for some  $i \in \mathbb{N}$ ,  $d(x, x_i) < \delta_x$ . Now,

$$x \in B_d(x_i, \delta_x) \subset B_d(x, 2\delta_x) \subset B_d(x, r) \subset U$$

Consequently, we shown that:

$$\forall x \in U, \exists B_x \in \beta \text{ s.t. } x \in B_x \subset U$$

Now, Construct given  $U \in \mathcal{T}$  as: Since **Axiom of Choice**, Define  $\mathcal{C} \stackrel{\text{def}}{=} \bigcup_{x \in U} \{B_x \in \beta \mid x \in B_x \subset U\}$ . Then,

$$U = \bigcup_{u \in \mathcal{C}} u$$

□

**Theorem 0.0.5.** If Second-Countable, then Separable.

*Proof.* Let  $\beta = \{B_i \mid i \in \mathbb{N}, B_i \neq \emptyset\}$  be a Countable Basis of  $X$ .

By the **Axiom of Choice**, we can construct a countable set  $A = \{x_i \mid i \in \mathbb{N}, x_i \in B_i\}$ . WTS:  $\bar{A} \supset X$ .

Let  $x \in X$  and  $U \in \mathcal{T}$  with  $x \in U$  be given arbitrarily.

Since  $\beta$  is basis of  $\mathcal{T}$ ,  $x \in B_i \subset U$  for some  $B_i \in \beta$ . Thus,  $x_i \in U$ , so we obtain  $A \cap U \neq \emptyset$ . Therefore,  $x \in \bar{A}$ . □

**Theorem 0.0.6.** If Second-Countable, then First-Countable.

*Proof.* Trivial. □